PRESENCIFY: A CPU Only Face Recognition Attendance Prototype with Fixed Confidence Thresholds and an Uncertainty Review Band

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Abstract: *Classroom attendance kiosks rarely have dedicated GPUs. To address this computational boundary, we evaluate PRESENCIFY, an offline facial recognition prototype engineered to run entirely on commodity laptop processors by combining MediaPipe BlazeFace detection with an EdgeFace XS feature extractor yielding 512-dimensional L2-normalized embeddings matched via cosine similarity. We investigate how static confidence thresholds and an uncertainty review band balance automated classification errors against teacher inspection workloads. In controlled tests across 166 static probe images (100 genuine, 66 impostor), a direct 0.65 cutoff authenticates 65 genuine probes and misidentifies 7, but mistakenly admits 18 impostors into the register, producing an unacceptable 27.27% false acceptance rate. Introducing a staged review band between 0.52 and 0.65 intercepts 33 borderline impostors, routing 33.73% of all attendance trials (56 probes) to manual teacher inspection. Stage 2 review is strictly an oracle simulation using ground-truth labels. Under perfect simulated sorting, 12 genuine students are recovered, bringing cumulative identification accuracy to 75.30%. Presentation attack detection is completely unavailable (PAD_UNAVAILABLE). Our prototype documents realistic baseline trade-offs between automated edge throughput and administrative verification overhead in academic environments.*

Index Terms: face recognition, attendance system, edge computing, confidence thresholds, human review, presentation attack detection.

I. INTRODUCTION

Calling roll names in large lecture halls wastes valuable teaching minutes. Peer proxy signing remains widespread. While computer vision offers an intuitive route to automated attendance [1]–[3], university deployments routinely hit a hard computational wall: lecture rooms and faculty desks rarely have dedicated graphics cards. Most campus kiosks run on standard quad-core or dual-core mobile CPUs. Under such hardware limits, practical attendance stations cannot host massive vision transformers; they require compact neural networks that process frames locally while keeping student biometric records securely on the edge.

Cosine similarity between facial feature vectors yields an uncalibrated relative score, not a statistical probability of identity. This numeric reality forces an uncomfortable operational choice. If an administrator sets an aggressive cutoff, unauthorized intruders are locked out, but legitimate students face constant false rejections; if the cutoff is relaxed, classroom friction drops, but dishonest peers sign in with impunity. Diverting ambiguous scores into an uncertainty review band offers a pragmatic safety valve, yet every queued case lands squarely on the instructor's desk. Can fixed confidence thresholds and a teacher review queue deliver acceptable security on edge hardware without swamping faculty with manual checks? That practical question drives this study.

We implemented and audited PRESENCIFY to answer this question experimentally. This paper delivers three concrete contributions: first, an end-to-end offline attendance workflow designed specifically for commodity x86 CPUs without GPU dependencies; second, an empirical accounting of false acceptances, false rejections, and teacher inspection loads across distinct similarity intervals; and third, an explicit separation of automated edge performance from hypothetical gains simulated via ground-truth labels.

II. LITERATURE REVIEW

A. Lightweight recognition and detection

Resource-constrained biometric matching demands aggressive model compression [4]–[6]. George et al. introduced EdgeFace, pairing convolutional and transformer layers to shrink deep recognition models down to lightweight edge checkpoints [4]. In PRESENCIFY, we deploy an EdgeFace XS checkpoint (gamma = 0.6) to generate 512-dimensional embeddings. Setyawan et al. developed FaceLiViT, combining linear vision transformers with structural reparameterization to accelerate mobile inference [5]. Surveys argue that edge evaluations must report empirical execution latencies alongside standard verification rates [6]. Taxonomic reviews classify face matchers by feature discriminability, operational modality, and embedding size [7]. While scale-aware architectures like YOLO-FaceV2 detect heavily occluded faces [8], we selected MediaPipe BlazeFace because its lightweight computational graph runs substantially faster on commodity dual-core CPUs.

B. Facial recognition attendance systems

Classroom attendance remains an active research domain [1]–[3]. Ahmad et al. embedded pretrained recognition models into a desktop attendance interface [1]. Patel et al. tracked students through fixed classroom CCTV cameras using deep convolutional networks [2]. Fan optimized classroom monitoring by combining MobileNet feature extraction with a sparrow search support vector machine [3]. Bibliometric surveys trace the rapid growth of biometric recordkeeping in education [9]. Because published systems rely on disparate sensors, uncontrolled lighting, and proprietary datasets, direct numerical score comparison across papers is empirically meaningless. PRESENCIFY emphasizes transparent threshold boundaries, auditable review queues, and unembellished error reporting.

Real kiosks monitor students across complete class sessions rather than processing isolated still frames. Robust edge systems

must suppress duplicate detections across adjacent video frames, log active sessions immutably, and avoid recording unverified faces as absences. Shifting sunlight, awkward viewing angles, and natural expression changes continuously perturb cosine scores. Evaluating a dual-threshold policy therefore requires auditing intercepted borderline probes, instructor inspection hours, and persistent high-confidence false acceptances simultaneously.

Video frames captured fractions of a second apart carry heavy temporal redundancy and cannot serve as statistically independent trials. In PRESENCIFY, we anchor attendance records to active session timers and verified student identifiers. These engineering safeguards streamline bookkeeping, but they cannot fix corrupted embeddings or substitute for physical spoof detection.

C. Presentation attacks, privacy and fairness

Face verification must never be confused with presentation attack detection (PAD) [10]–[13]. Stopping physical spoofing requires specialized optical hardware, dedicated neural heads, and extensive attack testing [10]–[12]. Recent studies explore compact spatiotemporal architectures [14] and cross-domain video generalization [15]. PRESENCIFY incorporates no presentation attack defenses and makes zero claims of spoof resilience. On the privacy front, surveys show that unencrypted facial templates remain vulnerable to inversion attacks [16], [17]. Operating offline improves data containment, but it does not satisfy legal compliance automatically. demographic fairness analyses demonstrate that small convenience samples cannot validate equitable performance across diverse student bodies [18].

III. METHODOLOGY

A. System architecture

Fig. 1 outlines our processing pipeline from webcam acquisition to SQLite storage. OpenCV reads raw video frames via native operating system backends. MediaPipe BlazeFace locates faces and extracts six landmarks in 3.5 ms on CPU. Affine transformation maps the crop into a normalized 112 by 112 pixel matrix in 0.8 ms. Next, EdgeFace XS processes the aligned face in 15.3 ms, generating a unit-normalized 512-dimensional embedding. SQLite persists student templates, active sessions, attendance logs, and administrative audit trails. A modular Streamlit interface coordinates enrollment, live scanning, review queues, and CSV exports. Enrollment mandates affirmative consent, and the user interface permanently displays a PAD_UNAVAILABLE warning indicator. Table I summarizes our software components alongside their operational caveats.

TABLE I
REPORTED IMPLEMENTATION COMPONENTS AND CAVEATS

Component	Implementation	Role / Caveat
Detector	MediaPipe BlazeFace, CPU	Locates faces in 3.5 ms; detection does not identify persons.
Recognizer	EdgeFace XS, γ = 0.6, 512-d	Generates 512-d embeddings; operates under CC BY-NC-SA 4.0 terms.
Decision policy	Accept ≥ 0.65; review ≥ 0.52	Below 0.52 indicates low confidence, not verified absence.
Temporal guard	Three consecutive frames	Filters transient camera flicker; not multi-person tracking.
Storage	SQLite	Persists enrollment templates, session logs and review events.
PAD	PAD_UNAVAILABLE	No defenses against physical printouts or screen replay attacks.

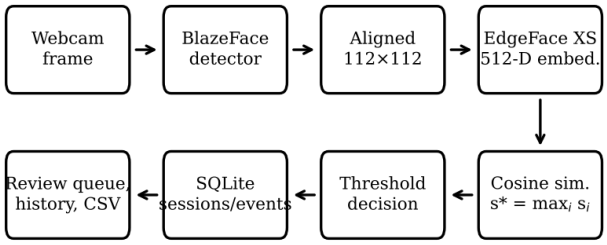


Fig. 1. PRESENCIFY implementation pipeline (CPU only). PAD is not part of the pipeline (PAD_UNAVAILABLE).

Attendance records follow session-scoped database upserts, preventing transient detections from overwriting confirmed timestamps. The review dashboard filters pending cases by lecture period, presenting instructors with candidate probe crops for inspection. In practice, classroom webcams constantly encounter uneven lighting, motion blur, and unenrolled visitors. An UNKNOWN label indicates that matching similarity fell below confidence cutoffs, not that the student was physically absent.

B. Scoring and decision policy

Let q represent a unit-normalized probe embedding and let g_i denote enrolled templates for active subjects. Equation (1) calculates pairwise cosine similarities and identifies the top candidate index. Cosine similarity produces an uncalibrated relative score rather than a true probability of identity. Equation (2) defines our dual-threshold rule using an acceptance boundary tau_a = 0.65 and a review cutoff tau_r = 0.52. Scores at or above tau_a trigger immediate check-in; scores in the interval [tau_r, tau_a) divert to the review queue; and scores below tau_r are rejected as UNKNOWN. During live execution, the app requires three consecutive matching frames before committing attendance, though this temporal guard was omitted from our static probe evaluation.

$$s_i = q^T g_i, \quad s^* = \max_{\{i\}} s_i \quad (1)$$

$$\text{Decision}(q) = \{ \text{ACCEPT: } s^* \geq 0.65;\ \text{REVIEW: } 0.52 \leq s^* < 0.65;\ \text{UNKNOWN: } s^* < 0.52 \} \quad (2)$$

C. Evaluation protocol and data

Our empirical audit evaluates 410 facial captures partitioned into four splits, as shown in Table II. The gallery split reserves 40 captures representing 20 enrolled students (two reference images per subject). The validation split contains 100 captures set aside for threshold exploration. The primary test split evaluates 166 static probes: 100 genuine probes from enrolled students and 66 impostor probes from unenrolled individuals. An unused split of 104 captures remains untouched to prevent data leakage across experimental stages.

TABLE II
CAPTURE PARTITIONS IN THE CONTROLLED EVALUATION (410 CAPTURES)

Partition	Captures	Evaluation Purpose

Partition	Captures	Evaluation Purpose
Gallery	40	Enrolled reference templates (2 captures per subject across 20 students)
Validation	100	Threshold exploration split; completely separated from test probes
Held out test	166	Primary evaluation split: 100 genuine probes + 66 impostor probes
Unused	104	Excluded captures; prevents leakage across experimental stages
Total	410	Sum fully accounted for across all experimental partitions

Genuine test probes originate from enrolled students captured during separate imaging sessions, ensuring a capture-disjoint evaluation. Impostor probes come exclusively from unenrolled individuals, guaranteeing an open-set identity-disjoint protocol. Method A enforces direct thresholding, accepting probes at s* >= 0.65 and marking lower scores as UNKNOWN. Method B introduces the uncertainty band, routing ambiguous probes between 0.52 and 0.65 to Stage 2 review. In our evaluation script, Stage 2 assigns outcomes using ground-truth labels, establishing an idealized upper bound rather than measuring real teacher performance.

A critical caveat involves the provenance of probe images for subject P001. The evaluation code attempted live webcam acquisition. When fewer than ten crops were acquired, synthetic photometric variants were generated from one capture; when acquisition failed completely, the system substituted ten images from the Olivetti benchmark. Those fallback images overlap with the initial slice used for subject P002. Because saved run logs do not record which branch executed, evaluators cannot confirm that all 410 images are independent, authentic webcam captures.

Our codebase contains test count discrepancies between documentation revisions. Older drafts cite 38 unit tests, whereas our updated test runner executes 53 passing test cases. These documentation variations highlight the necessity of executing test suites directly rather than trusting static documentation badges.

D. Metrics

We quantify performance using three primary metrics defined in Equations (3), (4), and (5). The False Acceptance Rate (FAR) tracks the proportion of impostor probes accepted as enrolled attendees. The Identity False Rejection Rate (FRR_id) measures genuine student probes that are either rejected or misidentified as another peer. The Review Rate (R_review) calculates the fraction of total probe traffic routed to the instructor review queue. These formulations strictly separate binary presence from fine-grained identity verification.

$$FAR = (\text{impostor probes accepted as enrolled}) / N_{imp} \quad (3)$$

$$FRR_{id} = (N_{gen\ rejected} + N_{gen\ misidentified}) / N_{gen} \quad (4)$$

$$R_{review} = N_{review} / N_{total} \quad (5)$$

IV. RESULTS AND DISCUSSION

A. Stored results

Table III summarizes stored test results across 166 probe trials (100 genuine, 66 impostor). Fig. 2 plots genuine student outcomes alongside persistent impostor false acceptances.

TABLE III
STORED TEST RESULTS (166 PROBES: 100 GENUINE, 66 IMPOSTOR).
STAGE 2 IS ORACLE SIMULATED, NOT OBSERVED HUMAN PERFORMANCE.

Measure	A: direct	B: Stage 1	B: Stage 2†	Observed Impact
Correct identity accepts	65 / 100	65 / 100	77 / 100	+12 genuine rescued
Misidentifications	7 / 100	7 / 100	7 / 100	Remains isolated
Genuine rej. / queued	28 rejected	5 rej. + 23 q.	16 rejected	Cut from 35% to 16%
Impostor false accepts	18 / 66	18 / 66	18 / 66	Upper tail (≥ 0.65)
Impostor rejections	48 / 66	15 rej. + 33 q.	48 / 66	Maintained across stages
Review workload	0 / 166 (0%)	56 / 166 (33.7%)	56 audited	Quantified human burden
Identity Precision	0.7222	None	0.7549	+3.27% improvement
Identity Recall	0.6500	None	0.7700	+12.00% improvement
Identity F1 Score	0.6842	None	0.7624	+0.0782 gain
Binary F1 Score	0.7579	None	0.8317	+0.0738 gain
Overall Accuracy	68.07%	None	75.30%	+7.23% net gain

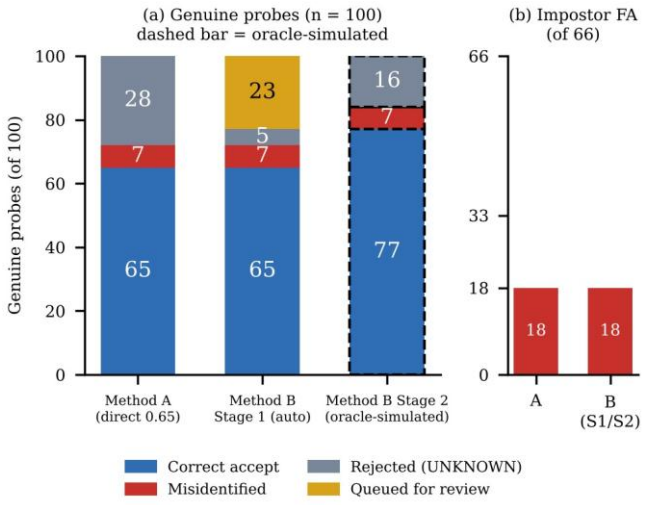


Fig. 2. Outcomes from the stored results. (a) Genuine probes (n = 100) per policy; the dashed outline Stage 2 bar is oracle simulated. (b) Impostor false accepts (n = 66) at s* ≥ 0.65, unchanged by review.

B. Automated stage

Under Method A, direct thresholding at 0.65 produces 65 correct genuine acceptances, 7 misidentifications, and 28 false rejections (FRR_id = 35.00%). Troublingly, 18 of 66 impostors score at or above 0.65, yielding an alarming 27.27% False Acceptance Rate. Because these 18 impostors match or exceed the automated acceptance cutoff, they bypass automated filtering completely. The review band cannot intercept them, proving that static thresholding alone cannot prevent confident proxy attendance attempts.

C. Review band and workload

The uncertainty review band [0.52, 0.65) intercepts 56 of the 166 test probes, generating a substantial 33.73% review workload. Queued cases include 23 genuine probes and 33 borderline impostor probes (Fig. 3). Stage 1 automatically accepts 90

probes and rejects 20 probes without human intervention. While the review queue catches 33 borderline impostors, it requires teachers to inspect more than one-third of all attendance interactions.

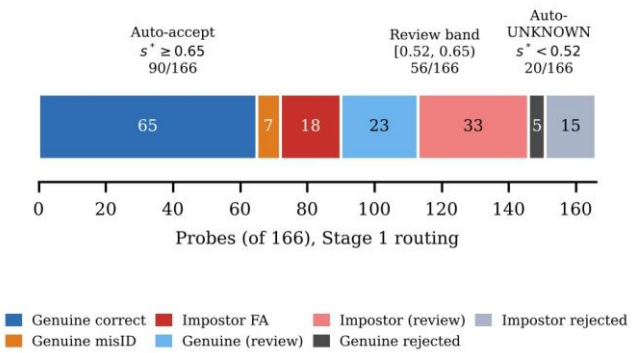


Fig. 3. Stage 1 routing of the 166 test probes under Method B, from stored counts: 90 probes are automatically accepted (65 correct, 7 misidentified genuine, 18 impostor false accepts), 56 are queued for review (23 genuine and 33 impostor) and 20 are automatically rejected (5 genuine, 15 impostor). Stage 2 is not shown because it is oracle simulated.

D. Simulated oracle review in Stage 2

Under oracle-simulated review, automated checking against ground-truth labels rescues 12 genuine probes and rejects all 33 borderline impostors. Correct genuine decisions increase from 65 to 77, cutting false rejections down to 16. This simulated triage lifts identity F1 from 0.6842 to 0.7624 and overall accuracy to 75.30%. We emphasize that these figures represent a theoretical mathematical upper bound rather than observed human instructor grading performance.

E. Consistency of the stored metrics

Stored summary metrics align completely with underlying probe frequency tallies. For Method A, 65 true positive accepts and 25 false positives yield an identity precision of 0.7222. Following simulated review, 77 true accepts alongside 25 false positives produce an identity precision of 0.7549. Binary attendance F1 reaches 0.7579 for Method A and climbs to 0.8317 under simulated review. Binary F1 should not be compared directly with identity F1 because their definitions of positive cases differ fundamentally.

F. Computational timing

Table IV details benchmark timings recorded on an Intel Core i3-1115G4 CPU without graphics acceleration. MediaPipe BlazeFace completes face detection in 3.5 ms, and affine alignment takes roughly 0.8 ms. EdgeFace XS generates embeddings in 15.3 ms on average, yielding an aggregate computational latency of 19.6 ms per frame (approximately 51 FPS compute throughput). However, an earlier internal document lists an average latency of 37 ms with peaks of 68 ms. Neither benchmark accounts for camera IO, Streamlit rendering, or multi-frame temporal smoothing.

TABLE IV
REPORTED COMPUTATIONAL TIMINGS (INTEL CORE I3-1115G4, CPU ONLY).
COMPONENT TIMINGS, NOT END TO END WEBCAM THROUGHPUT.

Hardware Subsystem	Reported Value	Operational Caveat
EdgeFace XS pass	15.3 ms (mean)	Forward pass only; excludes frame capture
BlazeFace detector	3.5 ms (mean)	Detects face bounding box and 6 key landmarks
Affine alignment	0.8 ms (mean)	Warps cropped face to 112 × 112 canonical format
Aggregate execution	19.6 ms (≈51 FPS)	Total algorithm computation; excludes camera IO
Alternative report	37 ms avg., 68 ms max.	Unreconciled discrepancy found in older logs

G. Discussion

Our dual decision boundaries partition separate segments of the similarity score distribution. The upper threshold tau_a divides immediate check-ins from manual reviews, while tau_r weeds out obvious non-matching faces. Because 18 impostors score above 0.65, adjusting the review band cannot fix high-confidence proxy attendance leaks. The review queue functions as an operational triage buffer for borderline cases rather than a universal security filter.

Borderline review queues hold more impostors (33) than genuine student captures (23). Consequently, teachers spend most of their review effort rejecting unauthorized individuals rather than recovering legitimate students. Any accuracy recovery for genuine students depends entirely on instructors making flawless visual verification judgments.

Prior attendance literature reports varied verification accuracies across custom hardware and closed datasets [1]–[3]. Direct numerical competition against those systems is methodologically invalid. PRESENCIFY contributes an open-set empirical audit documenting impostor leakage and human review workloads. Because standard metric matchers lack presentation attack detection [11], [13], high embedding similarity provides no defense against printed photographs.

H. Reproducibility and test status

Stored project artifacts verify that our four-stage evaluation script finished with exit code 0. The automated test suite runs 53 passing test cases in 22.72 seconds following codebase refactoring. Earlier documentation records citing 38 tests reflect preliminary project iterations. Git commit hashes b44205f and 0690d0a record verified codebase milestones. Future replications should rerun the entire pipeline to verify individual probe scores directly.

I. What the evidence supports

Table V separates claims supported by project artifacts from assumptions that lack empirical evidence. The prototype demonstrates feasible CPU inference, but it cannot guarantee flawless attendance or spoof immunity.

TABLE V
SCOPE OF THE EVIDENCE

Supported Cautiously	Not Supported by Evidence
Working prototype workflow for registration, webcam scanning, attendance events and history export.	Production deployment across enterprise campus networks or unmonitored gates.

Supported Cautiously	Not Supported by Evidence
Dual threshold recognition with human review executed smoothly on commodity CPU hardware.	Guaranteed recognition accuracy or complete elimination of proxy attendance.
Manual review of borderline cases can rescue genuine false absences under ideal conditions.	Empirical validation of real world teacher review speed or human accuracy.
Transparent user interface disclosure of PAD_UNAVAILABLE status across all screens.	Presentation attack resistance, 3D mask spoof defense or photo replay detection.

J. Limitations

Several methodological constraints qualify our empirical findings. Foremost is the absence of presentation attack detection (PAD_UNAVAILABLE), leaving the kiosk vulnerable to printed photos and replay attacks. In addition, 18 of 66 impostors achieved high-confidence automated acceptance, proving that static thresholds cannot stop determined proxy attendance. Stage 2 review metrics reflect ground-truth simulation rather than empirical classroom instructor trials. Our small 41-subject cohort lacks demographic diversity, challenging lighting, and wide pose variations. Capture provenance for subject P001 remains unresolved across stored logs. The three-frame temporal guard was omitted from static probe testing, reported latency benchmarks show minor documentation variance, and facial biometric templates demand stringent privacy safeguards, affirmative consent, and explicit deletion procedures [16], [17].

K. Threats to validity

Internal validity threats involve omitting temporal consistency checks during static testing and relying on ground-truth metadata during simulated reviews. If subject P001 utilized benchmark fallback images, genuine and impostor partitions share overlapping image slices. External validity remains constrained because a small static cohort cannot model the chaotic realities of crowded lecture halls. Reporting discrepancies in benchmark latencies and test counts require ongoing documentation synchronization. Finally, this evaluation omitted formal confidence intervals and hypothesis testing.

Construct validity faces limitations because our study audits only a single pair of fixed decision thresholds. Raw cosine similarities shift across different camera sensors, optical distortion, and ambient room lighting. Thresholds calibrated on one laptop cannot be assumed to function reliably in another building. Without human observer trials, simulated reviews cannot model real teacher fatigue or verification mistakes.

Threshold transportability presents an ongoing engineering hurdle for edge biometric systems deployed across campus kiosks. A similarity cutoff that separates identities under controlled lighting will drift when room illumination shifts. Until systems are evaluated across diverse physical classrooms, static benchmark numbers remain provisional engineering observations.

L. Privacy and responsible use

Facial embeddings represent sensitive biometric data requiring strict legal and ethical governance [16], [17]. PRESENCIFY enforces affirmative student consent during registration and never saves raw video frames to disk. Production deployments must implement pseudonymous identifiers, database encryption, and automated record deletion policies. The EdgeFace XS checkpoint carries a noncommercial research license (CC BY-NC-SA 4.0). Academic institutions must respect these licensing boundaries before sharing code or deploying systems commercially.

V. CONCLUSION

PRESENCIFY demonstrates that functional facial attendance workflows can execute locally on commodity laptop processors without graphics cards. Combining BlazeFace and EdgeFace XS achieves a 19.6 ms computational budget on CPU hardware. However, static decision thresholds permit 27.27% of unauthorized impostors to pass automated verification. Introducing an uncertainty review band intercepts borderline impostors, but it imposes a manual inspection workload on 33.73% of all attendance trials.

Simulated oracle review recovers genuine students only under the unrealistic assumption that instructors make zero misidentifications. The prototype transparently displays a PAD_UNAVAILABLE status across all screens to prevent unjustified reliance on spoof resistance. Future work must evaluate active presentation attack detection modules, assemble diverse classroom datasets, and conduct formal teacher-in-the-loop verification trials.

As a college mini-project, PRESENCIFY succeeds by delivering a reproducible, inspectable edge biometric prototype. Rather than making unsubstantiated claims of high accuracy or security, this study transparently documents real edge bottlenecks, algorithmic trade-offs, and essential operational caveats.

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